

* Find the central region of galaxies to sometimes output large amounts of energy.

→ Center "appears" pointlike due to resolution

• Optical spectra: bright nuclei cannot be explained in terms of "normal" mechanism such as:

(i) Supernovae

(ii) Nuclear fusion

→ Means we need a new mechanism to describe the central area of a galaxy.

* These galaxies: which have centers w/ high energy outputs are

Active Galactic Nuclei (AGN)

* AGN spectrum:

• Find emission in all parts of light spectrum
Radio $\leftrightarrow$ X-ray ...

• Cause of emission in each section due to different component of AGN (as we'll see).

* Grav: $E = mc^2$ * E_m of Grav.

$$\Delta U = -(U_f - U_i) = \frac{GM_R m}{R}$$

$$\Delta U = mc^2 \Rightarrow E = \frac{GM_{BH}}{Rc^2}$$

$$* m_e: U(R) = -\frac{GMm}{r} \quad \Bigg| \quad K = \frac{1}{2} m v_{rot}^2$$

$$\frac{GMm}{R^2} = \frac{v^2 m}{R} \Rightarrow v_{rot}^2 = \frac{GM_{BH}}{R}$$

$$K = \frac{1}{2} m G M_{BH} / R$$

$$\Delta U = \frac{dU}{dr} \Delta r = \frac{G M_{BH} m}{r^2} \Delta r$$

$$\Delta K = \left| \frac{dK}{dr} \right| \Delta r = \frac{G M_{BH} m}{2 r^2} \Delta r$$

$$\frac{\Delta U}{\Delta K} = 2 \Rightarrow \Delta K = \frac{\Delta U}{2}$$

50% : K.E.

50% : Radiative

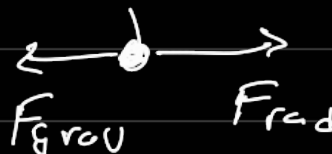
$\epsilon_{max} = 1/4$ disk

ϵ : radiative

$$L = \frac{dE}{dt} = \epsilon \frac{dM}{dt} c^2$$

$$L = \epsilon \dot{M} c^2$$

→ radiation: γ : $E_\gamma = pc$



$$F_{rad} = \frac{dp}{dt} = \frac{1}{c} \underbrace{\frac{dE}{dt}}_{\text{Luminosity}}$$

$$F_{rad} = \frac{L_\sigma}{c}$$

L_σ : Luminosity fraction

L_{tot} : total luminosity is out in all directions
however we will only get a fraction:

$$L_{\sigma} = \frac{\sigma_T}{4\pi R^2} L_{\text{tot}}$$

$$F_{\text{rad}} = \frac{\sigma_T L_{\text{tot}}}{4\pi R^2 c}$$

- Plasma moves $m_e + m_p$

$$F_{\text{grav}} = \frac{GM_{\text{BH}}(m_p + m_e)}{R^2}$$

- Stable orbit:

$$F_{\text{grav}} > F_{\text{rad}} \Rightarrow \frac{GM_{\text{BH}} m_p}{R^2} = \frac{\sigma_T L_{\text{tot}}}{4\pi R^2 c^2}$$

$$L_{\text{tot}} = \frac{4\pi GM_{\text{BH}} m_p c}{\sigma_T} \quad (\text{Eddington limit})$$

$\Rightarrow$ Minimum mass

Seyfert: $M_V \leq -23$

Quasars, Bright

Type 1: Both broad & narrow line emission

- Blue & UV bands

Type 2: narrow lines

$$L_1 > L_2$$

* Radio Jets: ~ 100 's kpc

FR 1/2: Radio loud

Magnetic energy in Lobes $\sim 10^{54}$ J

Radio: Synchrotron radiation of ultra rel.

X-ray: AGN: brightest point sources

Gal. Preferences:

Ellipticals: Quasars

Spirals:

(BLR): Broad-line region:

• dense clouds $n > 10^{10} \text{ cm}^{-3}$

→ doppler shifts of clusters near center

(NLR) Narrow-line region:

• Forbidden QM transitions

↳ low densities / Activity

• further from nucleus

• Few kpc across

BH: $M_{BH} \sim 10^6 - 10^{10} M_{\odot}$

• $R_{inf} \sim 10 \text{ pc}$

→ Duty cycle of an AGN: $K = \frac{\sigma_T}{m_H}$

$$F_{rad} = \frac{K}{c} \frac{L}{4\pi r^2} = \frac{\sigma_T L}{c m_H 4\pi r^2}$$

$$\begin{aligned} \therefore L_{Edd} &= \frac{4\pi G M_{BH} m_p}{\sigma_T} = 1.25 \times 10^{38} \left(\frac{M}{M_{\odot}} \right) \text{ erg s}^{-1} \\ &= 1.26 \times 10^{31} \left(\frac{M_{BH}}{M_{\odot}} \right) \text{ W} \end{aligned}$$

→ Maximum mass: BH

• If $L > L_{Edd}$: stop even more mass accretion

- If source has a large reservoir: tend towards Lead

$$\dot{M}_{\text{edd}} = \frac{4\pi G M m}{c \sigma_T c} M$$

$$M(t) = M_0 e^{t/t_{\text{edd}}}$$

$$t_{\text{edd}} \equiv \frac{M}{\dot{M}_{\text{edd}}} = \frac{c \sigma_T c}{4\pi G M m} \quad (\text{Eddington time})$$

t_{edd} : Typical timescale of a shining AGN

For $\epsilon \approx 0.1$

$$t_{\text{edd}} \approx 4.5 \times 10^7 \text{ yr}$$

* Nuclear: $\sim 3 \times 10^{10} M_\odot$

Supernova: $\sim 5 \times 10^{12} M_\odot$

SMBH: $\sim 10^6 - 10^{10} M_\odot$

→ Matter falls in ϵ , heats up: emits radiation

Recall:

$$\Delta U = \frac{1}{2} \Delta V$$

so other half is radiated.

* Consider a mass m : $\Gamma_{\text{min}} = c \left(\frac{G M_{\text{BH}}}{c^2} \right)$

$$\Gamma_{\text{min}} = c R_G$$

$$E \approx -\frac{1}{2} U = \frac{1}{2} \frac{G M m}{\Gamma_{\text{min}}} = \frac{1}{2} \frac{G M_{\text{BH}} m c^2}{c G M_{\text{BH}}} \\ = \frac{m c^2}{2 c}$$

$$E \approx m c^2 / 2 c \quad (E_n \text{ radiated away})$$

$$\eta \approx \frac{1}{2}$$

$$efficiency \approx \epsilon \approx z_a$$

$$a = 3-6 \quad so \quad \epsilon \approx .2 - .2$$

* Gravitational redshift: $r > R_s$

$$z = \frac{1}{\sqrt{1 - R_s/r}} - 1$$

$$z = (\lambda - \lambda_e) / \lambda_e$$

$$E_\gamma = hc / \lambda \quad | "m" = E_\gamma / c^2$$

$$-\lambda_e + \frac{\lambda_e}{\sqrt{1 - R_s/r}} = \lambda - \lambda_e$$

$$\lambda = \frac{\lambda_e}{\sqrt{1 - (R_s/r)}}$$

$$E_\gamma = \frac{hc}{\lambda} = \frac{hc}{\lambda_e} \sqrt{1 - \left(\frac{R_s}{r}\right)} \approx \frac{hc}{\lambda_e} \left(1 - \frac{1}{2} \frac{R_s}{r}\right)$$

$$E_\gamma = \frac{hc}{\lambda_e} - \frac{GMm}{r}$$

Gas Disk Kinematics:

measure vel. of gas in Disk

$$\text{Kep. orbits:} \quad M = \frac{RV^2}{G}$$

Stellar Kinematics:

Broad X-Ray Lines:

• X-ray em. at 6.4 eV from Fe Fluorescence

→ suggests this emission is very close to center

Reverberation mapping: DLR

$$\bullet \text{Time delay:} \quad R_{BLR} \approx c \Delta t$$

$$V_{BLR}: \quad f R_{BLR} \frac{V_{DLR}}{c}$$

$$M_{BH} \approx$$

BH - Galaxy Relations:

$$\frac{M_{BH}}{M_{\odot}} \approx \left(\frac{1}{1000} \right) \left(\frac{L_{bulge}}{L_{\odot}} \right)^{1.1}$$

$$\frac{M_{BH}}{M_{\odot}} \approx 2 \times 10^8 \left(\frac{\sigma_{bulge}}{200 \text{ km/s}} \right)^{4.9}$$

σ_{bulge} : stellar velocity dispersion

* BLR: $\lesssim 0.01 \text{ pc}$

- Gas clouds in Quick motion $\sim 10^4 \text{ km/s}$
- high density & T

Toroidal Region: Gas & Dust

- $\sim 1 \text{ pc}$

NLR: $\sim 100 \text{ pc}$

- warm clouds
- slower motion & density

